ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q3, Q4

Question 1

(a) f_x = 3200 px. FOV = 160 mm. 1 pixel = 0.125 mm.

(b) Answer: 0.122 N.

(c) ZOH discretisation: $F = e^{AT}$, $G = \int_0^T e^{A\tau} B\, d\tau = A^{-1}(F - I)B$ $(A \neq 0)$.

Here $A = -2$, $B = 1$, $T = 0.1$: $F = e^{-0.2} = \mathbf{0.8187}$. $G = (F - 1)/A = (0.8187 - 1)/(-2) = \mathbf{0.0906}$.

$x_{k+1} = 0.8187\, x_k + 0.0906\, u_k$.

Question 3

(a) (1) Differentiation is a high-pass operation (gain $\propto \omega$), so encoder measurement/quantisaton noise is amplified, by about $1/T = 1000$ at 1 kHz.

(2) Encoder quantisaton: $\Delta\theta$ is a whole number of counts, so $\hat{\omega} = \Delta\theta/T$ moves in steps of (count size)/T; at low speed the estimate is a coarse staircase of zeros and spikes.

(b) $y[k] = 0.0591\,(u[k] + u[k-1]) + 0.8818\, y[k-1]$. DC gain = 1.

(c) $L = [18, 64]^T$.

(d) Advantage of the obsever: it uses the plant model and the applied input u, so its velocity estimat has (almost) no phase lag; the 20 Hz filter on differenced data always delays by about $1/(2\pi f_c) \approx 8$ ms. It can also reconstruct states that cannot be obtained by differencing (e.g. disturbance, when augmented).

Filter is safer when the model is uncertain or changes (unknown payload, friction, contact, actuator saturation): the obsever then gives a biased or poorly-behaved estimate while the model-free filter still follows the measured data.

Question 4

(a) Poles: $-2 \pm 2j$. $N_{ff} = 8$.

(b) $\theta_{ss} = 0.8$, error = 20%.

(c) $k_1 = 18$, $k_2 = 6$, $K_I = 32$.

(d) For a constant r, in steady state all derivatives vanish: $\dot{\xi} = r - \theta_{ss} = 0 \Rightarrow \theta_{ss} = r$, independent of b and c; the closed loop only needs to be stabel so that a steady state exists.

Failure mode: integrator windup. When the actuator saturates the error persists, ξ keeps growing and causes a large overshoot afterwards. Remidy: anti-windup (clamp the integrator, conditional integration or back-calculation).